Transformada de Laplace: Para solucionar EDP.

Transformada integral.

Definición 1: Sea $f(t)$ una función definida para $t \geq 0$ entonces la integral impropia:

$$\int_0^{\infty} K(s,t)\, f(t)\, dt = \lim_{b \to \infty} \int_0^{b} K(s,t)\, f(t)\, dt,$$

existe.

Vamos a seleccionar $K(s,t) = e^{-st}$. Así, se obtiene la Transformada de Laplace.

Definición 2: Transformada de Laplace.

Sea $f(t)$ una función definida $t \geq 0$ Entonces la integral

$$\int_0^{\infty} e^{-st} f(t)\, dt = \lim_{b \to \infty} \int_0^{b} e^{-st} f(t)\, dt$$

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} f(t)\, dt,$$

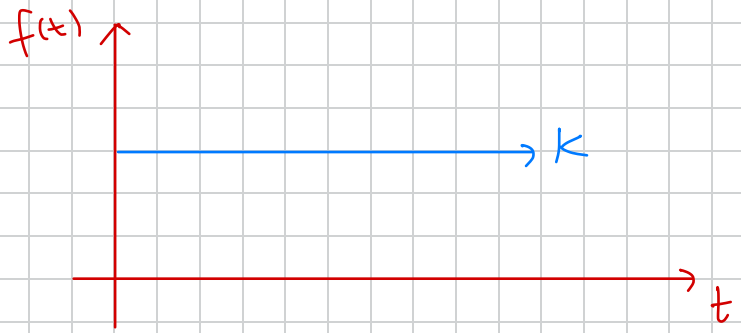
es la Transformada de Laplace de f(t), toda vez que la integral sea convergente.

Dominio del tiempo t $\longrightarrow$ Dominio de la frecuencia $S. = j2\pi f$

$$\mathcal{L}\{f(t)\} = F(s) \quad ; \quad \mathcal{L}\{g(t)\} = G(s) \quad ;$$

$$\mathcal{L}\{y(t)\} = Y(s).$$

Ejemplo: Calcular $\mathcal{L}\{f(t)\}$ si $f(t) = K \in \mathbb{R}$ es una constante en los reales para todo $t \geq 0$.



Vamos aplicar la definición 2:

$$\int_0^{\infty} e^{-st} f(t)\,dt = \lim_{b\to\infty} \int_0^{b} e^{-st} f(t)\,dt$$

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} f(t)\,dt,$$

$$\mathcal{L}\{K\} = \int_0^{\infty} e^{-st}(K)\,dt = K\int_0^{\infty} e^{-st}\,dt$$

$$= \lim_{b\to\infty} \int_0^{b} e^{-st}\,dt = K \lim_{b\to\infty} -\frac{e^{-st}}{s}\Big|_0^{b}$$

$$= K\left[\lim_{b\to\infty} -\frac{e^{-sb}}{s} + \frac{1}{s}\right] = \frac{K}{s}.$$

($e^{-sb} \to 0$)

$$\mathcal{L}\{K\} = F(s) \quad ; \quad \mathcal{L}\{K\} = \frac{K}{s}$$

Transformada de Laplace de una constante.

$\mathcal{L}\{t\}$

Vamos aplicar la definición 2:

$$\int_0^\infty e^{-st} f(t)\,dt = \lim_{b\to\infty} \int_0^b e^{-st} f(t)\,dt$$

$$\mathcal{L}\{t\} = \int_0^\infty e^{-st}\, t\, dt, \quad \text{IPP.}$$

$\int$		$\int$
t	$+$	e^{-st}
1	$-$	$\frac{e^{-st}}{s}$
0	$+$	$\frac{e^{-st}}{s^2}$

$$= \lim_{b\to\infty} \left(-\frac{t}{s} e^{-st} - \frac{e^{-st}}{s^2} \right)\Bigg|_0^b$$

$$= \lim_{b\to\infty} -\frac{b e^{-sb}}{s} - \frac{e^{-sb}}{s^2} + \frac{1}{s^2} = \frac{1}{s^2}$$

$$\mathcal{L}\{t\} = \frac{1}{s^2}$$

Propiedad de linealidad: Sean $f(t)$ y $g(t)$ dos funciones definidas $t \geq 0$, Sean α y $\beta \in \mathbb{R}$.

$$\mathcal{L}\{\alpha f(t) \pm \beta g(t)\} = \mathcal{L}\{\alpha f(t)\} \pm \mathcal{L}\{\beta g(t)\}$$
$$= \alpha \mathcal{L}\{f(t)\} \pm \beta \mathcal{L}\{g(t)\}$$
$$= \alpha F(s) \pm \beta G(s).$$

Definición 3: Transformada de Laplace de algunas funciones básicas.

$$\mathcal{L}\{k\} = \frac{k}{s}$$

$$\mathcal{L}\{t\} = \frac{1}{s}$$

$$\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}} \quad ; \quad n = 1, 2, 3, \ldots$$

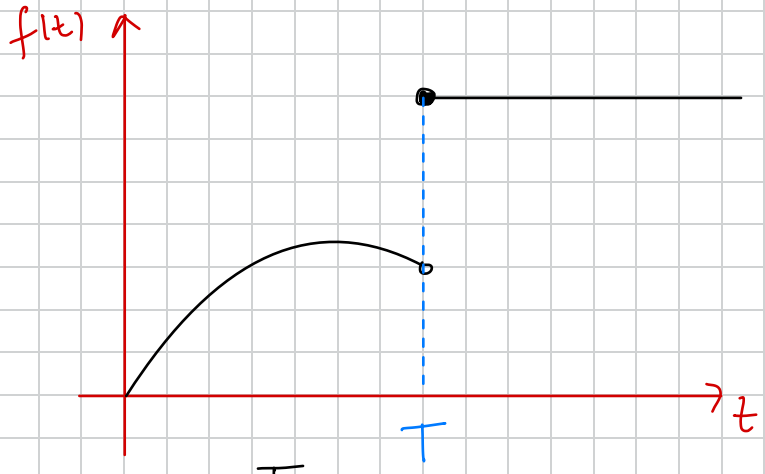
$$\mathcal{L}\{e^{at}\} = \frac{1}{s-a}$$

$$\mathcal{L}\{\operatorname{sen}(wt)\} = \frac{w}{s^2+w^2}$$

$$\mathcal{L}\{\cos(wt)\} = \frac{s}{s^2+w^2}$$

$$\mathcal{L}\{\operatorname{senh}(wt)\} = \frac{w}{s^2-w^2}$$

$$\mathcal{L}\{\cosh(wt)\} = \frac{s}{s^2-w^2}$$



$$\mathcal{L}\{f(t)\} = \int_0^T e^{-st} f(t)\,dt + \int_T^\infty e^{-st} f(t)\,dt$$

Ejemplo:

$$f(t) = \begin{cases} 2t+1 & ; \quad 0 \le t < \pi \\ \operatorname{Sen}(t) & ; \quad t \ge \pi \end{cases}$$

$$\mathcal{L}\{f(t)\} = \int_0^\pi e^{-st}(2t+1)\,dt + \int_\pi^\infty e^{-st}\operatorname{Sen}(t)\,dt$$

$\int$ IPP $\int$

$2t+1$	$+$	e^{-st}	
		$\searrow$	$-\frac{e^{-st}}{s}$
2	$-$	$\searrow$	$\frac{e^{-st}}{s^2}$
0	$+$	$\int$	$\rightarrow$

$$(2t+1)\left(-\frac{e^{-st}}{s}\right) - \frac{2e^{-st}}{s^2}\Bigg|_0^\pi$$

$$(2\pi+1)\left(-\frac{e^{-s\pi}}{s}\right) - 2\frac{e^{-s\pi}}{s^2} - \left(-\frac{1}{s} - \frac{2}{s^2}\right) =$$

$$-(2\pi+1)\frac{e^{-s\pi}}{s} - \frac{2e^{-s\pi}}{s^2} + \frac{1}{s} + \frac{2}{s^2} =$$

$$\frac{\left(1 - (2\pi+1)e^{-s\pi}\right)}{s} + \frac{2}{s^2}\left(1 - e^{-s\pi}\right)$$

$$\int_\pi^\infty e^{-st}\,\mathrm{sen}(t)\,dt = \left(-\mathrm{sen}(t)\frac{e^{-st}}{s} - \frac{\cos(t)\,e^{-st}}{s^2}\right)\Big|_\pi^\infty$$

$\int$ IPP $\int$ $-\int_\pi^\infty \frac{e^{-st}}{s^2}\,\mathrm{sen}(t)\,dt$

$\mathrm{sen}(t)$	$+$	e^{-st}
$\cos(t)$	$-$	$-e^{-st}/s$
$-\mathrm{sen}(t)$	$+\int$	e^{-st}/s^2

$$\left(1+\frac{1}{s^2}\right)\int_\pi^\infty e^{-st}\,\mathrm{sen}(t)\,dt = \lim_{b\to\infty}\left(-\mathrm{sen}(t)\frac{e^{-st}}{s} - \frac{\cos(t)\,e^{-st}}{s^2}\right)\Big|_\pi^\infty$$

$$= \lim_{b\to\infty} -\frac{\mathrm{sen}(b)\,e^{-sb}}{s} - \frac{\cos(b)\,e^{-sb}}{s^2} - \frac{e^{-s\pi}}{s^2}$$

$$\int_\pi^\infty e^{-st}\,\mathrm{sen}(t)\,dt = \frac{-\frac{e^{-s\pi}}{s^2}}{1+1/s^2} = \frac{-e^{-s\pi}}{s^2+1}$$

($1+1/s^2 = \frac{s^2+1}{s^2}$)

$$= -\frac{e^{-s\pi}}{s^2+1}$$

$$\frac{\left(1-(2\pi+1)\,e^{-s\pi}\right)}{s} + \frac{2}{s^2}\left(1-e^{-s\pi}\right) - \frac{e^{-s\pi}}{s^2+1}$$

Definición 4: Transformada de Laplace inversa

Si $F(s)$ es la transformada de laplace de una función $f(t)$, es decir, $\mathcal{L}\{f(t)\} = F(s)$, entonces $f(t)$ es la transformada inversa de $F(s)$, así $f(t) = \mathcal{L}^{-1}\{F(s)\}$

$$t \xrightarrow{\mathcal{L}\{\}} s$$

$$s \xrightarrow{\mathcal{L}^{-1}\{\}} t$$

$$k = \mathcal{L}^{-1}\left\{\frac{k}{s}\right\} \; ; \; t = \mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} \; ; \; e^{-3t} = \mathcal{L}^{-1}\left\{\frac{1}{s+3}\right\}$$

Definición 5: Transformada de Laplace inversa de algunas funciones básicas.

$$k = \mathcal{L}^{-1}\left\{\frac{k}{s}\right\}$$

$$t^n = \mathcal{L}^{-1}\left\{\frac{n!}{s^{n+1}}\right\} \; ; \; n = 1, 2, 3 \ldots$$

$$\operatorname{sen}(\omega t) = \mathcal{L}^{-1}\left\{\frac{\omega}{s^2+\omega^2}\right\}$$

$$\cos(\omega t) = \mathcal{L}^{-1}\left\{\frac{s}{s^2+\omega^2}\right\}$$

$$e^{at} = \mathcal{L}^{-1}\left\{\frac{1}{s-a}\right\}$$

$$\operatorname{senh}(\omega t) = \mathcal{L}^{-1}\left\{\frac{\omega}{s^2-\omega^2}\right\}$$

$$\cosh(\omega t) = \mathcal{L}^{-1}\left\{\frac{s}{s^2-\omega^2}\right\}$$

Definición 6: Transformación de una derivada.

$$\int_0^{\infty} e^{-st} f'(t)\,dt = \lim_{b\to\infty} \int_0^{b} e^{-st} f'(t)\,dt$$

$$\mathcal{L}\{f'(t)\} = \int_0^{\infty} e^{-st} f'(t)\,dt$$

$$= \lim_{b\to\infty} \int_0^{b} e^{-st} f'(t)\,dt = \lim_{b\to\infty} f(t)e^{-st}\Big|_0^{b}$$

$$\overset{\text{IPP}}{\int} \qquad + s\underbrace{\int_0^{b} e^{-st} f(t)\,dt}_{\mathcal{L}\{f(t)\} = F(s)}$$

$$e^{-st} \quad (+) \quad f'$$

$$-se^{-st} \quad (-) \quad f$$

$$= \lim_{b\to\infty} f(b)e^{-sb} - f(0) + sF(s)$$

$$\mathcal{L}\{f'(t)\} = sF(s) - f(0).$$

$$\mathcal{L}\{f''(t)\} = \int_0^\infty e^{-st} f''(t)\,dt$$

$$\lim_{b\to\infty} \int_0^b e^{-st} f'(t)\,dt = \lim_{b\to\infty} f'(t)e^{-st}\Big|_0^b$$

IBP

$$\int e^{-st} \quad \int f''$$

$$e^{-st} \;(+) \;\to\; f'$$

$$-se^{-st} \;(-) \;\to\; f$$

$$s^2 e^{-st} \;(+) \;\to\; \int f$$

$$+ fse^{-st}\Big|_0^b + s^2\int_0^b e^{-st} f(t)\,dt$$

$$\mathcal{L}\{f(t)\} = F(s)$$

$$\mathcal{L}\{f''(t)\} = s^2F(s) - sf(0) - f'(0)$$

$$\mathcal{L}\{f^{(n)}(t)\} = s^nF(s) - s^{n-1}f(0) - s^{n-2}f'(0)$$

$$+ \cdots - f^{(n-1)}(0)$$

Transformada de Laplace de derivadas parciales.

La solución $u(x,t)$

$$\mathcal{L}\{u(x,t)\} = \int_0^\infty e^{-st} u(x,t)\,dt = U(x,s)$$

$$\mathcal{L}\left\{\frac{\partial u}{\partial t}\right\} = sU(x,s) - u(x,0)$$

$$\mathcal{L}\left\{\frac{\partial^2 u}{\partial t^2}\right\} = s^2 U(x,s) - s\,u(x,0) - \frac{\partial u}{\partial t}(x,0)$$

$$\mathcal{L}\left\{\frac{\partial u}{\partial x}\right\} = \lim_{b\to\infty} \int_0^\infty e^{-st} \frac{\partial u}{\partial x}\,dt$$

$$= \frac{\partial}{\partial x}\left(\lim_{b\to\infty} \int_0^\infty e^{-st} u(x,t)\,dt\right)$$

$$\mathcal{L}\{u(x,t)\}$$

$$= \frac{\partial}{\partial x}\mathcal{L}\{u(x,t)\}$$

$$= \frac{\partial U(x,s)}{\partial x} = \frac{dU(x,s)}{dx}$$

$$\mathcal{L}\left\{\frac{\partial^2 u}{\partial x^2}\right\} = \lim_{b\to\infty} \int_0^\infty e^{-st} \frac{\partial^2 u}{\partial x^2}\,dt$$

$$= \frac{\partial^2}{\partial x^2}\left(\lim_{b\to\infty} \int_0^\infty e^{-st} u(x,t)\,dt\right)$$

$$\mathcal{L}\{u(x,t)\}$$

$$= \frac{\partial^2}{\partial x}\mathcal{L}\{u(x,t)\}$$

$$= \frac{\partial^2 U(x,s)}{\partial x^2} = \frac{d^2 U(x,s)}{dx^2}$$

Ejemplo: $\frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}$; $0 < x < 1$

$0 < t < 1/2$

$u(x,0) = 0$, $u(1,t) = 0$; $0 \leq t \leq 1/2$

$u(x,0) = \text{Sen}(\pi x)$; $0 \leq x \leq 1$

$$\mathcal{L}\left\{\frac{\partial^2 u}{\partial x^2}\right\} = \mathcal{L}\left\{\frac{\partial u}{\partial t}\right\}$$

$$\frac{d^2 U(x,s)}{dx^2} = sU(x,s) - u(x,0)$$ Transformada de Laplace

$$\frac{d^2 U(x,s)}{dx^2} - sU(x,s) = \underbrace{-u(x,0)}_{-\text{Sen}(\pi x)}$$

EDO Segundo orden.

$$U(x,s) = U_h(x,s) + U_p(x,s)$$

$$\frac{d^2 U(x,s)}{dx^2} - sU(x,s) = 0$$

$$m^2 - s = 0$$

$$(m - \sqrt{s})(m + \sqrt{s}) = 0$$

$$U_h(x,s) = c_1 e^{-\sqrt{s}x} + c_2 e^{\sqrt{s}x}$$

coeficientes indeterminados

$$U_p(x,s) = A\cos(\pi x) + B\,\text{Sen}(\pi x)$$

$$\frac{dU_p}{dx} = -A\pi\,\text{Sen}(\pi x) + B\pi\cos(\pi x)$$

$$\frac{d^2 U_p}{dx^2} = -A\pi^2\cos(\pi x) - B\pi^2\,\text{Sen}(\pi x)$$

$$-A\pi^2\cos(\pi x) - B\pi^2\operatorname{sen}(\pi x)$$

$$-s(A\cos(\pi x) + B\operatorname{sen}(\pi x)) = -\operatorname{sen}(\pi x)$$

$$-A\pi^2\cos(\pi x) + sA\cos(\pi x) = 0$$

$$-B\pi^2\operatorname{sen}(\pi x) - sB\operatorname{sen}(\pi x) = -\operatorname{sen}(\pi x)$$

$$\cos(\pi x)(-A\pi^2 + sA) = 0 \quad ; \quad A = 0$$

$$B\operatorname{sen}(\pi x)(\pi^2 + s) = -\operatorname{sen}(\pi x)$$

$$B = \frac{-1}{s+\pi^2}$$

$$U(x,s) = c_1 e^{-\sqrt{s}x} + c_2 e^{\sqrt{s}x} - \frac{1}{s+\pi^2}\operatorname{sen}(\pi x)$$

$$U(x,0) = 0$$

$$0 = c_1 e^{-\sqrt{0}x} + c_2 e^{\sqrt{0}x} - \frac{1}{s+\pi^2}\operatorname{sen}(\pi x)$$

$$U(1,s) = c_1 e^{-\sqrt{s}} + c_2 e^{\sqrt{s}} = 0$$

$$\begin{bmatrix} 1 & 1 \\ e^{-\sqrt{s}} & e^{\sqrt{s}} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} \frac{1}{s+\pi^2}\operatorname{sen}(\pi x) \\ 0 \end{bmatrix}$$

$$\det = e^{\sqrt{s}} - e^{-\sqrt{s}}$$

$$c_1 = \frac{e^{\sqrt{s}}\operatorname{sen}(\pi x)}{s+\pi^2} \Big/ \det \quad ; \quad c_2 = \frac{-e^{-\sqrt{s}}\operatorname{sen}(\pi x)}{s+\pi^2} \Big/ \det$$

$$U(x,s) = c_1 e^{-\sqrt{s}x} + c_2 e^{\sqrt{s}x} - \frac{1}{s+\pi^2}\operatorname{Sen}(\pi x)$$

$$c_1 = \frac{\dfrac{e^{\sqrt{s}}\operatorname{sen}(\pi x)}{s+\pi^2}}{\det} \quad ; \quad c_2 = \frac{\dfrac{-e^{-\sqrt{s}}\operatorname{Sen}(\pi x)}{s+\pi^2}}{\det}$$

$$\det = e^{\sqrt{s}} - e^{-\sqrt{s}} = e^{\sqrt{s}} - \frac{1}{e^{\sqrt{s}}} = \frac{e^{s}-1}{e^{\sqrt{s}}}$$

$$c_1 \frac{\dfrac{e^{\sqrt{s}}\operatorname{sen}(\pi x)}{s+\pi^2}}{\dfrac{e^{s}-1}{e^{\sqrt{s}}}} = \frac{\operatorname{Sen}(\pi x)\, e^{s}}{(s+\pi^2)(e^{s}-1)}$$

$$c_2 = \frac{-\operatorname{Sen}(\pi x)e^{-s}}{(s+\pi^2)(e^{s}-1)}$$

$$U(x,s) = \frac{\operatorname{Sen}(\pi x)\, e^{-s^{3/2}x}}{(s+\pi^2)(e^{s}-1)} - \frac{\operatorname{Sen}(\pi x)e^{-s^{3/2}x}}{(s+\pi^2)(e^{s}-1)}$$

$$- \frac{1}{s+\pi^2}\operatorname{Sen}(\pi x)$$

$$\mathcal{L}^{-1}\{U(x,s)\} = u(x,t)$$

$$\mathcal{L}^{-1}\left\{ - \frac{1}{s+\pi^2}\operatorname{Sen}(\pi x) \right\} = -\operatorname{Sen}(\pi x)\, e^{-\pi t}$$

Solución:

$$u(x,t) = -\operatorname{Sen}(\pi x)\, e^{-\pi t}$$